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#SCT_DS_3
 # Decision Tree Classifier

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Create Sample Dataset
X, y = make_classification(
    n_samples=1000,
    n_features=10,
    n_informative=6,
    n_redundant=2,
    random_state=42
)

feature_names = [f"Feature_{i+1}" for i in range(10)]

df = pd.DataFrame(X, columns=feature_names)
df["Target"] = y

print(df.head())

# Split Data
X = df.drop("Target", axis=1)
y = df["Target"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Model
model = DecisionTreeClassifier(max_depth=5, random_state=42)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"\nAccuracy: {accuracy*100:.2f}%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Feature Importance
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

plt.figure(figsize=(10,5))
importance.plot(kind="bar")
plt.title("Feature Importance")
plt.tight_layout()
plt.show()

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

print("Task Completed Successfully!")
```

```
   Feature_1  Feature_2  Feature_3  Feature_4  Feature_5  Feature_6  \
0  -1.030931   1.391626   0.547274   0.928932  -1.738880   1.250002
1  -2.766254   1.247870  -0.303691   1.083145   0.710836   1.968202
2  -0.558987   0.299849   1.527071   0.360442  -1.360209   1.100793
3  -1.350289  -2.046078  -0.614264   0.126459  -0.783923   5.895026
4  -0.275754  -0.728495   0.027727  -0.660834  -1.928161   3.544945

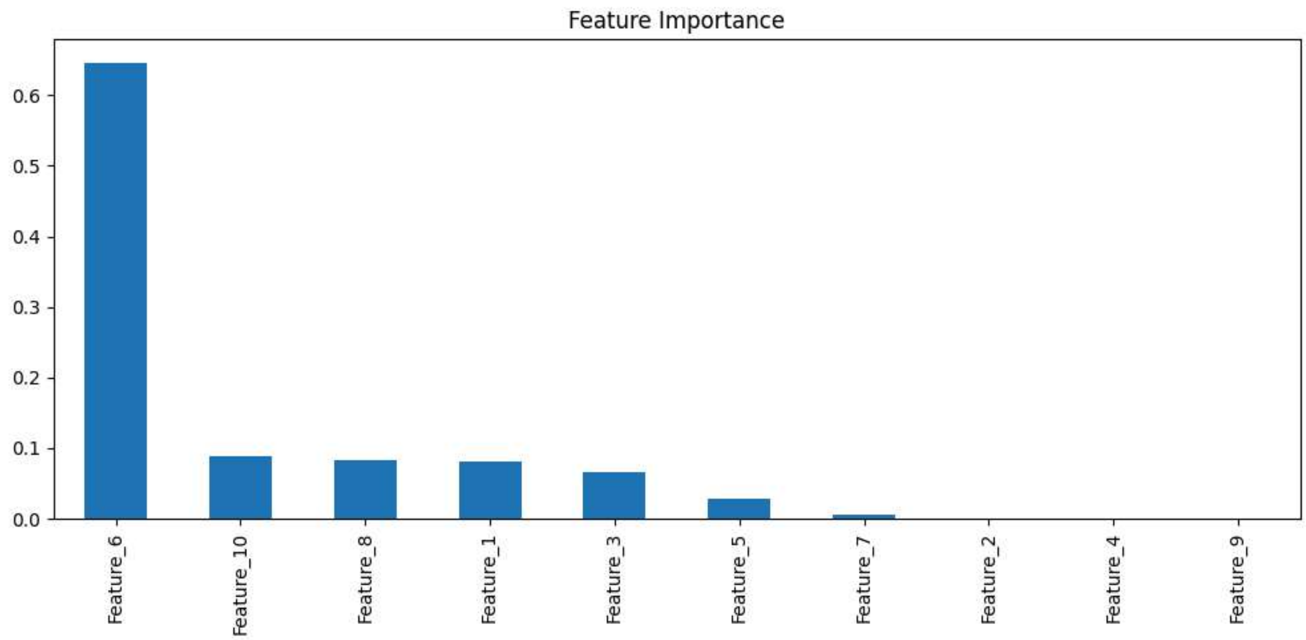
   Feature_7  Feature_8  Feature_9  Feature_10  Target
0   1.332551   1.578256   2.124722   -0.318434       0
1  -1.794192   2.346422   1.700778   -0.001190       1
2  -0.755951   1.331933   2.041105   -0.824404       0
3  -0.915477  -3.184768  -0.399260   -3.920960       0
4   1.446944  -1.111662   0.313766   -2.376528       0

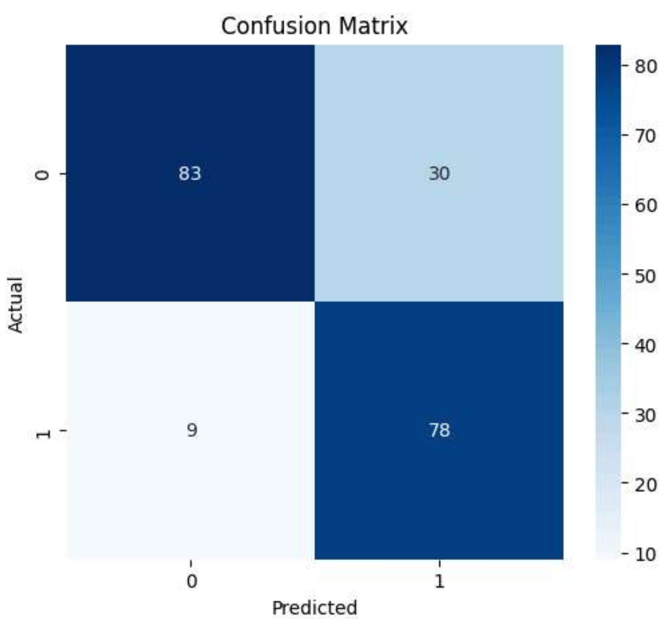
Accuracy: 80.50%

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.73      0.81       113
           1       0.72      0.90      0.80        87

    accuracy                           0.81       200
   macro avg       0.81      0.82      0.80       200
weighted avg       0.82      0.81      0.81       200
```



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Task Completed Successfully!
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